

Zhiwei Li

1997.08

PhD Student

AAII, University of Technology Sydney

15 Broadway, Ultimo NSW 2007, Australia

zev_ai

lizhw.cs@outlook.com

https://github.com/mtics

https://zhw.li/

Research Interests: On-device Personal Intelligence, Recommender Systems, Federated Learning

Education

- **University of Technology Sydney**[QS Top-100] — Sydney, Australia
  PhD, Computer Science, Faculty of Engineering and Information Technology, Supervisor: Guodong Long — 2023.11 - Present
- **ShanghaiTech University**[Double 1st-Class] — Shanghai, China
  Master's degree, Computer Science and Technology, School of Information Science and Technology, Supervisor: Lu Sun — 2020.09 - 2023.07
- **Zhengzhou University**[211, Double 1st-Class] — Zhengzhou, China
  Bachelor's degree, Computer Science and Technology, School of Information Engineering — 2016.09 - 2020.07

Project Experience

- **Building the Foundations of Federated Agentic AI** – ARC Future Fellowship, from 2027, CI: Guodong Long — 2026.03 - Present
  Policy consultation needs accurate answers while protecting user privacy. As a **participant in preliminary research**, I designed how AI assistants divide tasks and share information: search relevant policies by domain, use dialogue history to organize evidence, provide only necessary user information, and cite policy sources. In tests across several models, answer quality matched the centralized approach with about 35% fewer tokens per turn.
- **Federated Foundation Models for Recommendations** – ARC Discovery Project, CI: Guodong Long — 2025.08 - Present
  Combining images and text in the same way for everyone misses individual preferences. As a **research participant**, I designed a two-step recommendation process: create multiple combinations of images, text, and shared usage patterns, then adjust their weights using user records kept locally. In one comparison using the same base recommendation method, recommendation hit rate was about 19.7% higher than with a generic combination method.
- **Robust Federated Learning for Imperfect Decentralised Data** – ARC Discovery Project, CI: Chengqi Zhang — 2023.11 - 2025.03
  Scattered user data and limited feedback make it difficult to capture individual preferences. As a **research participant**, I designed two recommendation methods: one learns shared information and personal preferences separately while reducing the model information exchanged; the other adjusts the balance between the two using user records to predict preferences for content with no recorded feedback. For both, tests with individual parts removed or fixed checked what each part contributed. In one test, the second method's mean ranking score was about 15.6% higher than the strongest other federated method tested.

Selected Publications

- **Zhiwei Li**, Guodong Long, Jing Jiang, Chengqi Zhang, Qiang Yang, "Federated Vision-Language-Recommendation with Personalized Fusion," in AAAI, 2026. [CCF-A] [CORE-A*]
- **Zhiwei Li**, Guodong Long, Chunxu Zhang, Honglei Zhang, Chengqi Zhang, Jing Jiang, "A Survey of Personalized Federated Foundation Models for Privacy-Preserving Recommendation," in IJCAI, 2026. [CCF-A] [CORE-A*]
- **Zhiwei Li**, Guodong Long, Tianyi Zhou, Jing Jiang, Chengqi Zhang, "Personalized Federated Collaborative Filtering: A Variational AutoEncoder Approach," in AAAI, 2025. [CCF-A] [CORE-A*]
- **Zhiwei Li**, Guodong Long, Tianyi Zhou, "Federated Recommendation with Additive Personalization," in ICLR, 2024. [CCF-A] [CORE-A*]
- **Zhiwei Li**, Zijian Yang, Lu Sun, Mineichi Kudo, Keigo Kimura, "Incomplete Multi-View Weak-Label Learning with Noisy Features and Imbalanced Labels," in PRICAI, 2023. [CCF-C] [CORE-B]
- Honglei Zhang, **Zhiwei Li**, Haoxuan Li, Xin Zhou, Jie Zhang, Yidong Li, "TransFR: Transferable Federated Recommendation with Adapter Tuning on Pre-trained Language Models," in AAAI, 2026. [CCF-A] [CORE-A*]
- Chunxu Zhang, Guodong Long, Zijian Zhang, **Zhiwei Li**, Honglei Zhang, Qiang Yang, Bo Yang, "Personalized Recommendation Models in Federated Settings: A Survey," in IEEE TKDE, 2025. [CCF-A] [CORE-A*]
- Zijian Yang, **Zhiwei Li**, Lu Sun, "Generalized Discriminative Deep Non-Negative Matrix Factorization Based on Latent Feature and Basis Learning," in IJCAI, 2023. [CCF-A] [CORE-A*]

Professional Experience

- **Tutor,** Advanced Artificial Intelligence, University of Technology Sydney (Instructor: Jing Jiang) — 2026.08 - 2026.11
- **Research Assistant,** ShanghaiTech University (Supervisor: Lu Sun) — 2023.07 - 2023.11
- **Teaching Assistant,** Machine Learning, ShanghaiTech University (Instructor: Sibei Yang) — 2022.09 - 2023.01
- **Teaching Assistant,** Machine Learning, ShanghaiTech University (Instructor: Hao Wang) — 2022.02 - 2022.06

Selected Honors & Awards

- **AISTATS Best Reviewer Award** 2025.04
- **UTS AAII Student Best Paper Award** 2024.11
- **Merit Student, ShanghaiTech University** 2022.12
- **RoboCup Asia-Pacific Tianjin Invitational Tournament**, 2nd Prize 2019.05
- **RoboCup China Open Standard Platform League (NAO)**, 3rd Prize 2019.04
- **Third-Class Scholarship, Zhengzhou University** 2018.11

Academic Services

Reviewer / Program Committee Member:

- **Conferences**: NeurIPS, ICML, ICLR, KDD, WWW, AAAI, IJCAI, AISTATS, ECAI
- **Journals**: IEEE TPAMI, IEEE TKDE, IEEE TNNLS, ACM TOIS, IEEE TAI

Invited Speaker: WWW (2025), AJCAI (2024)

Languages

Mandarin (Native), English (IELTS: 6.5)